Basic part 2 (venv setup) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint builds toward the full solution.

Where this exercise is saved: `practice/venv_setup_practice.md` — notes, not a script. Every block below is a terminal command, so there is nothing to `python`-run: keep a Markdown file with each command you ran and the output you got back, and you will have your own record of the sequence the next time you start a project.

- Hint 1 — The idea, and the exact pieces
- Hint 2 — The plan, and almost the whole thing

Hint 1 — The idea, and the exact pieces

The mental model: `python -m venv .venv` creates an isolated Python environment (its own `site-packages` folder, separate from your system Python). Activating it (`source .venv/bin/activate`) changes which `python` and `pip` your shell resolves to, for the rest of that terminal session. From that point, anything you `pip install` only affects this project, not your whole system or any other project.

`pip freeze > requirements.txt` writes every installed package and its exact version to a file — this is what lets anyone (including future you) recreate the identical environment later with `pip install -r requirements.txt`.

Same four commands, with what to check after each one:

1. `python -m venv .venv` — check: a new `.venv/` folder now exists.
2. `source .venv/bin/activate` — check: your terminal prompt now shows `(.venv)` at the start.
3. `pip install requests` — check: `pip list` shows `requests` in the installed packages.
4. `pip freeze > requirements.txt` — check: opening `requirements.txt` shows a line like `requests==2.32.3`.

If step 2 doesn't seem to do anything visible, that's the most common place this goes wrong — see the Solution for what to check.

Hint 2 — The plan, and almost the whole thing

```
in your project folder:
    python -m venv .venv
    source .venv/bin/activate      (or .venv\Scripts\activate on Windows)
    pip install requests
    pip freeze > requirements.txt

confirm:
    prompt shows (.venv)
    requirements.txt contains a line starting with "requests=="
```

The commands, plus how to check each result:

```
# → practice/venv_setup_practice.md
python -m venv .venv
ls .venv                      # should show bin/, lib/, etc.

source .venv/bin/activate
which python                  # should now point inside .venv/

pip install requests
pip show requests             # confirms it installed correctly

pip freeze > requirements.txt
cat requirements.txt          # should list requests==<version>
```

Expected output of the last line, `cat requirements.txt`:

```
certifi==2024.8.30
charset-normalizer==3.3.2
idna==3.8
requests==2.32.3
urllib3==2.2.3
```

(Exact version numbers will differ depending on when you run this — `pip freeze` also lists `requests`'s own dependencies, not just `requests` itself, which is why you see 5 lines instead of 1.)

If activation seems to silently do nothing, see the Solution for the most common cause.

Full solution: Show me the solution